

Riemannian Manifolds

Labix

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Abstract

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1 Riemannian Metrics

1.1 The Riemannian Metric

Definition 1.1.1 (Riemannian Metric)

Let M be a smooth manifold. A Riemannian metric on M is a smooth section

$$g \in \Gamma^\infty((TM \otimes TM)^*)$$

such that $g_p : T_p(M) \times T_p(M) \rightarrow \mathbb{R}$ is an inner product for all $p \in M$.

Equivalently, a Riemannian metric is an assignment $p \mapsto g_p$ such that for any smooth sections $s, t \in \Gamma(M, TM)$, the map $p \mapsto g_p(s(p), t(p))$ is smooth.

Definition 1.1.2 (Riemannian Manifold)

A Riemannian manifold (M, g) is a manifold M together with a Riemannian metric g on M .

Given a (p, q) -tensor field F of a smooth vector bundle, recall that at a local frame $(U, s_1, \dots, s_k)$ the components of F are the smooth functions $F_{j_1, \dots, j_q}^{i_1, \dots, i_p}$ such that

$$F = \sum_{1 \leq i_1, \dots, i_p, j_1, \dots, j_q \leq k} F_{j_1, \dots, j_q}^{i_1, \dots, i_p} s_{i_1} \otimes \dots \otimes s_{i_p} \otimes s_{j_1}^* \otimes \dots \otimes s_{j_q}^*$$

locally. When $E = TM$, we can recover the components by substituting:

$$F_{j_1, \dots, j_q}^{i_1, \dots, i_p} = F \left(dx^{i_1}, \dots, dx^{i_p}, \frac{\partial}{\partial x^{j_1}}, \dots, \frac{\partial}{\partial x^{j_q}} \right)$$

Notice that g is a $(0, 2)$ -tensor field since $(TM \otimes TM)^* = TM^* \otimes TM^*$. So the components of g are given by

$$g_{ij} = g \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right)$$

In particular, for $p \in M$ and $v, w \in T_p M$, we have

$$g_p(v, w) = \sum_{1 \leq i, j \leq n} g_{i,j} \left(\frac{\partial}{\partial x^i} \Big|_p, \frac{\partial}{\partial x^j} \Big|_p \right) (dx^i)|_p(v) (dx^j)|_p(w)$$

Example 1.1.3

Define the metric g on S^n as follows. For $p \in S^n \subseteq \mathbb{R}^{n+1}$, we have

$$T_p S^n = \{v \in \mathbb{R}^{n+1} \mid \langle p, v \rangle_{\mathbb{R}^{n+1}} = 0\}$$

Then define

$$g_p(v, w) = \langle v, w \rangle_{\mathbb{R}^{n+1}}$$

Consider the charts $(U_+, \phi_+ = (x_1, \dots, x_n))$ and $(U_-, \phi_- = (x_1, \dots, x_n))$ where U_+ and U_- are the northern and southern hemispheres respectively. In terms of the coordinates in $\mathbb{R}^{n+1}$, the basis for the tangent space is given by pushing forward along the inverse of the coordinate charts, and this gives the basis

$$e_i \mp \frac{x_i}{1 - |x|^2} e_{n+1}$$

for $1 \leq i \leq n$ at the point $p \in \phi_\pm^{-1}(x)$ where $e_1, \dots, e_{n+1}$ is the standard basis vector of $\mathbb{R}^{n+1}$. Then we compute that

$$g_{ij} = \delta_{ij} + \frac{x_i x_j}{1 - |x|^2}$$

Proposition 1.1.4

Every smooth manifold admits a Riemannian metric.

1.2 Maps Preserving the Riemannian Metric**Proposition 1.2.1**

Let M, N be smooth manifolds. Let g be a Riemannian metric on N . Let $f : M \rightarrow N$ be a smooth map. Then f^*g is a Riemannian metric on M if and only if f is an immersion.

This shows that if we want a map $f : M \rightarrow N$ that preserves Riemannian metric, we need f to at least be an immersion.

Definition 1.2.2 (Isometric Embedding)

Let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Let $f : M_1 \rightarrow M_2$ be a smooth map. We say that f is an isometric embedding if the following are true.

- f is an immersion.
- $f^*g_2 = g_1$.

Theorem 1.2.3 (Nash)

Let (M, g) be a Riemannian manifold of dimension n . Then there exists an isometric embedding $f : M \rightarrow \mathbb{R}^{(n+2)(n+3)/2}$ with the standard Euclidean metric.

Definition 1.2.4 (Isometries)

Let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Let $f : M_1 \rightarrow M_2$ be an isometric embedding. We say that f is an isometry if there exists an isometric embedding $h : M_2 \rightarrow M_1$ such that

$$h \circ f = \text{id}_{M_1} \quad \text{and} \quad f \circ h = \text{id}_{M_2}$$

In this case, we say that M_1 and M_2 are isometric.

Proposition 1.2.5

Let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Let $f : M_1 \rightarrow M_2$ be a smooth map. Then the following are equivalent.

- f is an isometry.
- f is an isometric embedding and a diffeomorphism.
- f is an isometric embedding that is bijective.
- f is an isometric embedding that is surjective.

Definition 1.2.6 (Local Isometries)

Let (M, g) and (N, h) be two Riemannian manifolds. We say that they are locally isometric if for all $p \in M$, there exists an open neighbourhood $U \subseteq M$ of p and $V \subseteq N$ open and an isometry $f : U \rightarrow V$.

Definition 1.2.7 (Flat Manifolds)

Let (M, g) be a Riemannian manifold. We say that (M, g) is flat if it is locally isometric to $\mathbb{R}^n$ with the standard metric.

In general, not every Riemannian manifold is flat. This can be shown once we discuss curvatures and torsions. However, this is true when $n = 1$.

Lemma 1.2.8 Every 1 dimensional Riemannian manifold is flat.

1.3 The Musical Isomorphisms

Definition 1.3.1 (The Flat Map)

Let (M, g) be a Riemannian manifold. Let $p \in M$. For each $X \in T_p M$, define the flat map

$$\flat : T_p M \rightarrow T_p^* M$$

by sending $X \in T_p M$ to the map $X^\flat : T_p M \rightarrow \mathbb{R}$ by $X^\flat(Y) = g_p(X, Y)$.

Theorem 1.3.2 (The Musical Isomorphism)

Let (M, g) be a Riemannian manifold. Let $p \in M$. Then the flat map

$$\flat : T_p M \rightarrow T_p^* M$$

is an isomorphism.

Definition 1.3.3 (The Sharp Map)

Let (M, g) be a Riemannian manifold. Let $p \in M$. Define the sharp map

$$\sharp : T_p^* M \rightarrow T_p M$$

to be the inverse of the flat map.

Proposition 1.3.4

Let (M, g) be a Riemannian manifold. Then the following are true.

- The components of the flat map are given by g_{ij} .
- The components of the sharp map are given by elements of the inverse matrix $(g_{ij})^{-1}$.

We denote the components of the sharp map by g^{ij} .

1.4 The Induced Inner Product on Tensor Fields

Definition 1.4.1 (The Induced Metric on the Cotangent Bundle)

Let (M, g) be a Riemannian manifold. Define the induced metric on $T^* M$ to be the section $g^{-1} \in \Gamma(M, (T^* M \otimes T^* M)^*)$ by

$$g^{-1}(\alpha, \beta) = g(\alpha^\sharp, \beta^\sharp)$$

Proposition 1.4.2

Let (M, g) be a Riemannian manifold. Then the components of g^{-1} is given by the elements of the inverse matrix $(g_{ij})^{-1}$.

Definition 1.4.3 (Inner Product of Tensors)

Let (M, g) be a Riemannian manifold. Define the induced inner product on $\mathcal{T}_p^q(TM)$ as follows. For each $x \in M$ and $v_1, \dots, v_p, w_1, \dots, w_p \in T_x M$ and $\alpha_1, \dots, \alpha_q, \beta_1, \dots, \beta_q \in T_x^* M$, define

$$\langle v_1 \otimes \dots \otimes v_p \otimes \alpha_1 \otimes \dots \otimes \alpha_q, w_1 \otimes \dots \otimes w_p \otimes \beta_1 \otimes \dots \otimes \beta_q \rangle = \prod_{i=1}^p g_x(v_i, w_i) \cdot \prod_{j=1}^q g_x(\alpha_j, \beta_j)$$

Proposition 1.4.4

Let (M, g) be a Riemannian manifold. Then the components of the induced inner product on $\mathcal{T}_p^q(TM)$ is given by

$$\langle -, - \rangle_{i_1, \dots, i_p, j_1, \dots, j_q}^{k_1, \dots, k_p, l_1, \dots, l_q} = g_{i_1 k_1} \dots g_{i_p k_p} g^{j_1 l_1} \dots g^{j_q l_q}$$

1.5 Riemannian Manifolds as a Metric Space

Definition 1.5.1 (Length of a Tangent Vector)

Let (M, g) be a Riemannian manifold. Let $v \in T_p(M)$ be a tangent vector for $p \in M$. Define the length of v to be

$$|v|_g = \sqrt{g_p(v, v)}$$

Definition 1.5.2 (Piecewise Regular Paths)

Let (M, g) be a Riemannian manifold. A piecewise regular path in M consists of smooth functions $\gamma_i : [a_i, b_i] \rightarrow M$ for $1 \leq i \leq k$ such that the following are true.

- For $1 \leq i \leq k-1$, we have $\gamma_i(b_i) = \gamma_{i+1}(a_i)$.
- For $1 \leq i \leq k$, we have $|\gamma'_i(t)| \neq 0$ for all $t \in [a_i, b_i]$.

Definition 1.5.3 (Length of a Curve)

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a piecewise regular path in M . Define the length of γ to be

$$L(\gamma) = \int_a^b |\gamma'(t)|_g dt$$

Definition 1.5.4 (Riemannian Distance Function)

Let (M, g) be a Riemannian manifold. Define the Riemannian distance function of M to be the function $d_g : M \times M \rightarrow \mathbb{R}$ given by

$$d_g(p, q) = \inf \{ L(\gamma) \mid \gamma : I \rightarrow M \text{ is a piecewise regular path} \}$$

Proposition 1.5.5

Let (M, g) be a Riemannian manifold. Then the following are true.

- d_g is a metric on M .
- The topology of M is given by the metric topology induced by d_g .

Example 1.5.6

For a moment, take it for granted that the short distance between two points on S^n is just a segment of a great circle. Then for $p, q \in S^n$, we have

$$d(p, q) = \int_0^1 |\gamma'(t)|_g dt$$

where $\gamma : [0, 1] \rightarrow S^n$ is the path given as follows. Let $\theta \in [0, \pi]$ be the unique angle such that $\cos(\theta) = \langle p, q \rangle_{\mathbb{R}^{n+1}}$. Suppose that $p \neq \pm q$. Then γ is given by

$$\gamma(t) = \frac{\sin((1-t)\theta)}{\sin(\theta)} p + \frac{\sin(t\theta)}{\sin(\theta)} q$$

Then we have

$$\gamma'(t) = \theta \left(-\frac{\cos((1-t)\theta)}{\sin(\theta)} p + \frac{\cos(t\theta)}{\sin(\theta)} q \right)$$

Then we have $|\gamma'(t)| = \theta$ and so we have

$$d(p, q) = \theta$$

This is precisely the spherical metric introduced in Geometry.

2 Introduction to Connections

2.1 Connections on Vector Bundles

Recall that for a smooth vector bundle $p : E \rightarrow M$, we denote the space of smooth sections on E by $\Gamma(E)$. Moreover, $\mathfrak{X}(M)$ is the space of smooth sections on the tangent bundle TM .

Definition 2.1.1 (Connections)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. A connection on p is a map

$$\nabla : \mathfrak{X}(M) \times \Gamma^\infty(E) \rightarrow \Gamma^\infty(E)$$

where we denote $\nabla(V, T)$ by $\nabla_V(T)$, such that the following are true.

- $C^\infty(M)$ -linearity in first variable: For each $T \in \Gamma^\infty(E)$, the map $V \mapsto \nabla_V(T)$ is $C^\infty(M)$ -linear. This means that

$$\nabla_{fV+gW}(T) = f\nabla_V(T) + g\nabla_W(T)$$

for $V, W \in \mathfrak{X}(M)$, $f, g \in C^\infty(M)$.

- $\mathbb{R}$ -linearity in second variable: For each $V \in \mathfrak{X}(M)$, the map $T \mapsto \nabla_V(T)$ is $\mathbb{R}$ -linear. This means that

$$\nabla_V(\lambda T + \mu S) = \lambda \nabla_V(T) + \mu \nabla_V(S)$$

- Product rule: The map ∇ satisfies the following product rule:

$$\nabla_V(fT) = V(f) \cdot T + f\nabla_V(T)$$

Proposition 2.1.2

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Then E admits a connection.

Definition 2.1.3 (Set of Connections)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Define the set of connections of E to be

$$\mathcal{A}(E) = \{ \nabla : \mathfrak{X}(M) \times \Gamma^\infty(E) \rightarrow \Gamma^\infty(E) \mid \nabla \text{ is a connection} \}$$

Proposition 2.1.4

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. A connection ∇ on E is equivalently an $\mathbb{R}$ -linear map

$$\nabla : \Gamma^\infty(E) \rightarrow \Gamma^\infty(T^*M \otimes E) = \Omega^1(M) \otimes_{C^\infty(M)} \Gamma^\infty(E)$$

such that

$$\nabla(f \cdot s) = df \otimes s + f \cdot \nabla s$$

for all $f \in C^\infty(M)$ and $s \in \Gamma^\infty(E)$.

Definition 2.1.5 (The Connection Forms)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Let $(U, e_1, \dots, e_r)$ be a local frame of E . Define the connection 1-forms ω_j^i of ∇ at the frame by

$$\nabla_X(e_j) = \sum_{i=1}^r \omega_j^i(X) e_i$$

for all $X \in \mathfrak{X}(U)$.

Proposition 2.1.6

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection

on E . Let $(U, e_1, \dots, e_r)$ and $(U, \bar{e}_1, \dots, \bar{e}_r)$ be local frames of E . Suppose that ω_j^i and $\bar{\omega}_j^i$ are the corresponding connection forms. Suppose that $A : U \rightarrow \text{GL}(r, \mathbb{R})$ is a transition function such that $\bar{e} = Ae$. Then we have

$$\bar{\omega}_j^i = A^{-1}\omega_j^i A + A^{-1}dA$$

2.2 Connections of Tensor Fields

Definition 2.2.1 (The Induced Connection on the Dual Bundle)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on M . Define the induced connection on E^* to be the connection $\nabla : \mathfrak{X}(M) \times \Omega^1(M) \rightarrow \Omega^1(M)$ by

$$\nabla_X \xi(s) = X(\xi(s)) - \xi(\nabla_X s)$$

Definition 2.2.2 (The Induced Connection on Tensor Fields)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on M . Define the induced connection of ∇ to be the connection $\nabla : \mathfrak{X}(M) \times \mathcal{C}^\infty(\mathcal{T}_q^p(E)) \rightarrow \mathcal{C}^\infty(\mathcal{T}_q^p(E))$ given by

$$\begin{aligned} \nabla_X F(\omega_1, \dots, \omega_p, Y_1, \dots, Y_q) &= X(F(\omega_1, \dots, \omega_p, Y_1, \dots, Y_q)) \\ &\quad - \sum_{i=1}^p F(\omega_1, \dots, \nabla_X \omega_i, \dots, \omega_p, Y_1, \dots, Y_q) \\ &\quad - \sum_{j=1}^q F(\omega_1, \dots, \omega_p, Y_1, \dots, \nabla_X Y_j, \dots, Y_q) \end{aligned}$$

Definition 2.2.3 (The Total Covariant Derivative)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on M . Let F be a tensor field of type (p, q) . Define the total covariant derivative of F to be $\nabla F : X \mapsto \nabla_X F$.

2.3 The Curvature Tensor

Definition 2.3.1 (The Curvature Tensor)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on p . Define the curvature of ∇ to be the smooth section $R(\nabla) \in \mathcal{C}^\infty(\text{End}(E) \otimes \Lambda^2 T^*M)$ given by

$$R(\nabla)(X, Y)(s) = [\nabla_X, \nabla_Y](s) - \nabla_{[X, Y]}(s)$$

for $X, Y \in \mathfrak{X}(M)$ and $s \in \mathcal{C}^\infty(E)$.

Notice that the definition makes sense: Appealing to the tensor characterization lemma, a section of $\mathcal{C}^\infty(\text{End}(E) \otimes \Lambda^2 T^*M)$ corresponds to a homomorphism $\Lambda^2 TM \rightarrow \mathcal{C}^\infty(\text{End}(E))$. Applying the lemma to $\mathcal{C}^\infty(\text{End}(E)) = \mathcal{C}^\infty(E^* \otimes E)$ again, we see that the curvature tensor corresponds to a map

$$\Lambda^2 TM \rightarrow \text{Hom}_{\mathcal{C}^\infty(M)}(\mathcal{C}^\infty(E), \mathcal{C}^\infty(E))$$

Definition 2.3.2 (The Curvature Form)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Let $U \subseteq M$ be an open subset. Let $e_1, \dots, e_r$ be a local frame of E on U . Define the curvature forms of ∇ on U to be the unique two forms $\Omega_j^i \in \mathcal{C}^\infty(U, E)$ such that

$$R(\nabla)(X, Y)(e_j) = \sum_{i=1}^r \Omega_j^i(X, Y)e_i$$

for all $X, Y \in \mathfrak{X}$.

Theorem 2.3.3 (The Second Structural Equation)

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Let $U \subseteq M$ be an open subset. Let $e_1, \dots, e_r$ be a local frame of E on U . Then we have

$$\Omega_j^i = d\omega_j^i + \sum_{k=1}^r \omega_k^i \wedge \omega_j^k$$

Another way to think about it: The matrix $\Omega = (\Omega_j^i)$ is an element of $\Omega^2(U, \mathfrak{gl}(n, \mathbb{R}))$ (indeed locally $\text{End}(E)$ is isomorphic to $\mathfrak{gl}(n, \mathbb{R})$). Gluing along local patches give $\Omega = d\omega + \omega \wedge \omega$. One can prove that $R(X, Y) = \Omega(X, Y)$. Indeed, Ω comes from the coordinate description of R , so patching it up should recover R .

Proposition 2.3.4

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Let $U \subseteq M$ be an open subset. Let $e, \bar{e}$ be local frames of E on U . Suppose that $g : U \rightarrow \text{GL}(r, \mathbb{R})$ is a transition function such that $\bar{e} = ge$. Suppose that Ω_j^i and $\bar{\Omega}_j^i$ are the curvature forms of ∇ with respect to e and $\bar{e}$. Then we have

$$\bar{\Omega}_j^i = g^{-1} \Omega_j^i g$$

2.4 Directional Derivatives of a Vector Bundle**Definition 2.4.1 (Covariant Derivatives Along a Curve)**

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve. Let ∇ be a connection on E . Define the covariant derivative $D_t : \mathcal{C}^\infty(\gamma^*(E)) \rightarrow \mathcal{C}^\infty(\gamma^*(E))$ of γ to be the pullback of the connection ∇ on γ .

Proposition 2.4.2

Let M be a smooth manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve. Let ∇ be a connection on E . Then the covariant derivative is given in local charts $(U, \phi = (x^1, \dots, x^n))$ for $\gamma(t_0) \in M$ by the formula

$$(D_t V)_{\gamma(t_0)} = \sum_{i=1}^n \left(\frac{\partial a_i}{\partial x^i} \Big|_{\gamma(t_0)} \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} + a_i(\gamma(t_0)) \nabla \left(\gamma'(t_0), \frac{\partial}{\partial x^i} \Big|_{\gamma(t_0)} \right) \right)$$

where V is given locally on the same chart by

$$V_{\gamma(t)} = \sum_{i=1}^n a_i(\gamma(t)) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)}$$

Definition 2.4.3 (Parallel Sections)

Let M be a smooth manifold. Let $p : E \rightarrow B$ be a smooth vector bundle. Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve. Let ∇ be a connection on E . Define the parallel sections of γ along ∇ to be $\ker(D_t)$.

In other words, a parallel section is a local section $s : \text{im}(\gamma) \rightarrow E$ such that $D_t(s) = 0$.

Proposition 2.4.4

Let M be a smooth manifold. Let $p : E \rightarrow B$ be a smooth vector bundle. Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve. Let ∇ be a connection on E . Let $X \in E_{\gamma(0)}$. Then there exists a unique parallel section $s : [0, 1] \rightarrow E$ such that $s(0) = X$.

Definition 2.4.5 (Parallel Transports)

Let M be a smooth manifold. Let $p : E \rightarrow B$ be a smooth vector bundle. Let $\gamma : [0, 1] \rightarrow M$ be a

smooth curve. Let ∇ be a connection on E . Define the parallel transport map

$$P_\gamma : E_{\gamma(0)} \rightarrow E_{\gamma(1)}$$

by $X \mapsto s_X(1)$ where $s_X : [0, 1] \rightarrow E$ is the unique parallel section such that $s_X(0) = X$.

Lemma 2.4.6

Let M be a smooth manifold. Let $p : E \rightarrow B$ be a smooth vector bundle. Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve. Let ∇ be a connection on E . Then P_γ is a well defined linear map.

3 Connections on the Tangent Bundle

3.1 Affine Connections

Definition 3.1.1 (Affine Connections) Let M be a smooth manifold. An affine connection of M is a connection

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$$

on the tangent bundle TM .

The directional derivative is the canonical affine connection on $\mathbb{R}^n$. In fact, every smooth manifold has such a canonical connection that generalizes the directional derivative.

Since the tangent bundle has a canonical local frame for every local chart, affine connections admit canonical connection components.

Definition 3.1.2 (Connection Components)

Let M be a smooth manifold. Let ∇ be an affine connection. Define the connection components Γ_{ij}^k of ∇ locally at a chart $(U, x^1, \dots, x^n)$ of M by

$$\nabla_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} = \sum_{k=1}^n \Gamma_{ij}^k \frac{\partial}{\partial x^k}$$

for $1 \leq i, j, k \leq n$.

Proposition 3.1.3

Let M be a smooth manifold. Let ∇ be an affine connection. Let $X, Y \in \mathfrak{X}(M)$ be vector fields. Suppose that the components of X and Y are given by X^i and Y^i respectively. Then locally at a chart $(U, x^1, \dots, x^n)$ of M , we have

$$\nabla_X Y = \sum_{k=1}^n \left(X(Y^k) + \sum_{i,j=1}^n X^i Y^j \Gamma_{ij}^k \right) \frac{\partial}{\partial x^k}$$

Definition 3.1.4 (The Torsion Tensor)

Let M be a smooth manifold. Let ∇ be an affine connection. Define the torsion of ∇ to be the map $T(\nabla) : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ given by

$$T(\nabla)(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]$$

Proposition 3.1.5

Let M be a smooth manifold. Let ∇ be an affine connection. Then the components of the torsion tensor is given by

$$T(\nabla)_{ij}^k = \Gamma_{ij}^k - \Gamma_{ji}^k$$

Definition 3.1.6 (The Torsion Form)

Let M be a smooth manifold. Let ∇ be an affine connection. Let $(U, e_1, \dots, e_n)$ be a smooth local frame of E . Define the torsion form τ^i of ∇ at the frame by

$$T(\nabla)(X, Y) = \sum_{i=1}^n \tau^i(X, Y) e_i$$

Theorem 3.1.7 (The First Structural Equation)

Let M be a smooth manifold. Let ∇ be an affine connection. Let $(U, e_1, \dots, e_n)$ be a smooth local

frame of TM . Then we have

$$\tau^i = d\theta^i + \sum_{j=1}^n \omega_j^i \wedge \theta^j$$

where $(U, \theta^1, \dots, \theta^n)$ is the dual of the local frame in T^*M .

Theorem 3.1.8 (The First Bianchi Identity)

Let M be a smooth manifold. Let ∇ be an affine connection. Let $(U, e_1, \dots, e_n)$ be a smooth local frame of TM . Then we have

$$d\tau^i = \sum_{j=1}^n \Omega_j^i \wedge \theta^j - \omega_j^i \wedge \tau^j$$

where $(U, \theta^1, \dots, \theta^n)$ is the induced local frame of T^*M .

Theorem 3.1.9 (The Second Bianchi Identity)

Let M be a smooth manifold. Let ∇ be an affine connection. Let $(U, e_1, \dots, e_n)$ be a smooth local frame of TM . Then we have

$$d\Omega_j^i = \sum_{k=1}^n \Omega_k^i \wedge \omega_j^k - \omega_k^i \wedge \Omega_j^k$$

where $(U, \theta^1, \dots, \theta^n)$ is the induced local frame of T^*M .

Definition 3.1.10 (The Einstein Tensor)

Let M be a smooth manifold. Let ∇ be an affine connection. Let $(U, e_1, \dots, e_n)$ be a smooth local frame of TM . Define the Einstein tensor to be the $(0, 2)$ -tensor field whose components are given by

$$G_{ab} = R_{ab} - \frac{1}{2} g_{ab} R$$

Theorem 3.1.11 (The Contracted Second Bianchi Identity)

Let M be a smooth manifold. Let ∇ be an affine connection. Let $(U, e_1, \dots, e_n)$ be a smooth local frame of TM . Then we have

$$\sum_{a=1}^n \sum_{c=1}^n g^{ac} \nabla_{e_c} G_{ab} = 0$$

for all $1 \leq b \leq n$.

If ∇^{e_a} denotes raising the index ($\nabla^{e_a} = \sum_{c=1}^n g^{ac} \nabla_{e_c}$), then we recover the more common form of the Bianchi identity:

$$\nabla^{e_a} G_{ab} = 0$$

3.2 Curvature and Torsion of Affine Connections

Let M be a smooth manifold. For an affine connection ∇ , the definition implies that ∇ is a smooth section of

$$\text{End}(TM) \otimes \Lambda^2 T^*M \cong TM \otimes T^*M \otimes \Lambda^2 T^*M$$

and this is exactly a map $TM \times TM \times TM \rightarrow TM$ where we think of the first two inputs as antisymmetric.

Definition 3.2.1 (Curvature Components)

Let M be a smooth manifold. Let ∇ be an affine connection on M . Define the curvature components R_{kij}^l of ∇ locally at a chart $(U, x^1, \dots, x^n)$ of M by

$$R(\nabla) \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right) \frac{\partial}{\partial x^k} = \sum_{l=1}^n R_{kij}^l \frac{\partial}{\partial x^l}$$

Proposition 3.2.2

Let M be a smooth manifold. Let ∇ be an affine connection on M . Then locally at a chart $(U, x^1, \dots, x^n)$ of M , we have

$$R_{kij}^l = \frac{\partial \Gamma_{kj}^l}{\partial x^i} - \frac{\partial \Gamma_{ki}^l}{\partial x^j} + \sum_{m=1}^n (\Gamma_{mi}^l \Gamma_{kj}^m - \Gamma_{mj}^l \Gamma_{ki}^m)$$

Proposition 3.2.3

Let (M, g) be a Riemannian manifold. Then M is flat if and only if $R(\nabla^g) = 0$.

Sometimes it is more convenient to lower the last index.

Definition 3.2.4 (Riemannian Curvature Tensor)

Let M be a smooth manifold. Let ∇ be an affine connection on M . Define the Riemannian curvature tensor of ∇ by

$$Rm(\nabla) = R(\nabla)^b$$

Lemma 3.2.5

Let M be a smooth manifold. Let ∇ be an affine connection on M . Then the components of Rm is given by

$$Rm_{kijl} = \sum_{m=1}^n g_{lm} R_{kij}^m$$

3.3 Metric Connections

Definition 3.3.1 (The Lie Bracket on Smooth Vector Fields) Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Define the Lie bracket of X and Y to be the vector field $[X, Y]$ given by the formula

$$[X, Y]_p(f) = X(Y_p(f)) = Y(X_p(f))$$

for $p \in M$ and $f \in C_{M,p}^\infty$.

Proposition 3.3.2 Let M be a smooth manifold. Let $X, Y \in \mathfrak{X}(M)$ be smooth vector fields. Let (U, ϕ) be a chart on M . Let X and Y be given locally on the chart by the formula

$$X = \sum_{k=1}^n a_k \frac{\partial}{\partial x^k} \quad \text{and} \quad Y = \sum_{k=1}^n b_k \frac{\partial}{\partial x^k}$$

for $a_k, b_k : U \rightarrow \mathbb{R}$ smooth functions. Then the Lie bracket of X and Y is given locally on the chart by the formula

$$[X, Y] = \sum_{k=1}^n (X(b_k) - Y(a_k)) \frac{\partial}{\partial x^k} = \sum_{i=1}^n \left(\sum_{j=1}^n \left(a_j \frac{\partial b_i}{\partial x^j} - b_j \frac{\partial a_i}{\partial x^j} \right) \right) \frac{\partial}{\partial x^i}$$

Definition 3.3.3 (Metric Connections) Let M be a smooth manifold. Let ∇ be an affine connection. We say that ∇ is a metric connection if the

$$\nabla(X, g(Y, Z)) = g(\nabla(X, Y), Z) + g(Y, \nabla(X, Z))$$

for all $X, Y, Z \in \mathfrak{X}(M)$.

3.4 The Levi-Civita Connection

Definition 3.4.1 (Levi-Civita Connections)

Let (M, g) be a Riemannian manifold. Let $\nabla^g : \mathfrak{X}(M) \times \mathfrak{X}(M) \rightarrow \mathfrak{X}(M)$ be an affine connection on M . We say that ∇^g is a Levi-Civita connection if the following are true.

- ∇^g is a metric connection.
- For any $X, Y \in \mathfrak{X}(M)$, we have

$$\nabla^g(X, Y) - \nabla^g(Y, X) = [X, Y]$$

Theorem 3.4.2 (Existence and Uniqueness of Levi-Civita Connections) Let (M, g) be a Riemannian manifold. Then M has a unique Levi-Civita connection.

Definition 3.4.3 (Christoffel Symbols)

Let (M, g) be a Riemannian manifold. Define the Christoffel symbols of ∇ to be the connection coefficients of the Levi-Civita connection.

Proposition 3.4.4

Let (M, g) be a Riemannian manifold. Then locally at a chart $(U, x^1, \dots, x^n)$ of M we have

$$\Gamma_{ij}^k = \frac{1}{2} \sum_{l=1}^n g^{kl} \left(\frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right)$$

for $1 \leq i, j, k \leq n$.

Theorem 3.4.5 (The Decomposition Theorem)

3.5 Curvature and Torsion of the Levi-Civita Connection

Proposition 3.5.1

Let (M, g) be a Riemannian manifold. Then the following are true regarding $Rm(\nabla^g)$.

- $Rm(W, X, Y, Z) = -Rm(X, W, Y, Z)$.
- $Rm(W, X, Y, Z) = -Rm(W, X, Z, Y)$.
- $Rm(W, X, Y, Z) = Rm(Y, Z, W, X)$.

Corollary 3.5.2

Let (M, g) be a Riemannian manifold. Then the following are true regarding the components of $Rm(\nabla^g)$.

- $Rm_{kijl} = -Rm_{ikjl}$.
- $Rm_{kijl} = -Rm_{kilj}$.
- $Rm_{kijl} = Rm_{jlk i}$.

Proposition 3.5.3 (The Bianchi Identities)

Let (M, g) be a Riemannian manifold. Then the following are true.

- $Rm(W, X, Y, Z) + Rm(X, Y, W, Z) + Rm(Y, W, X, Z) = 0$.
- $\nabla Rm(X, Y, Z, V, W) + \nabla Rm(X, Y, V, W, Z) + \nabla Rm(X, Y, W, Z, V) = 0$.

3.6 Ricci and Scalar Curvature

Definition 3.6.1 (Ricci Curvature)

Let (M, g) be a Riemannian manifold. Let $p \in M$. Define the Ricci curvature of M to be the map $\text{Ric} : TM \times TM \rightarrow \mathbb{R}$ given by

$$\text{Ric}(X, Y) = \text{tr}(Z \mapsto R(\nabla^g)(Z, X)(Y))$$

Definition 3.6.2 (Ricci Components)

Let (M, g) be a Riemannian manifold. Define the Ricci components R_{ij} locally at a chart $(U, x^1, \dots, x^n)$ of M by

$$R_{ij} = \text{Ric} \left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j} \right)$$

Proposition 3.6.3

Let (M, g) be a Riemannian manifold. Then locally at a chart $(U, x^1, \dots, x^n)$ of M , we have

$$R_{ij} = \sum_{k=1}^n R_{kij}^k$$

Definition 3.6.4 (Scalar Curvature)

Let (M, g) be a Riemannian manifold. Define the scalar curvature of M by

$$R = \sum_{i,j=1}^n g^{ij} R_{ij}$$

4 Geometry on Riemannian Manifolds

4.1 Geodesics on Riemannian Manifolds

Let M be a smooth manifold. Let $\gamma : I \rightarrow M$ be a smooth curve on M . Then we can compute its differential to obtain a tangent vector

$$\gamma'(t) \in T_{\gamma(t)}M$$

for each $t \in I$. This defines a vector field in $\mathfrak{X}(\gamma)$.

Definition 4.1.1 (Geodesics) Let M be a smooth manifold. Let ∇ be an affine connection on M . A geodesic on M is a curve $\gamma : I \rightarrow M$ such that

$$D_t(\gamma'(t)) = 0$$

where D_t is the associated covariant derivative of ∇ and γ .

We have seen geodesics for metric spaces, but these are slightly different. We require geodesics on manifolds to only minimize the distance locally.

4.2 Minimizing Curves

Definition 4.2.1 (Minimizing Curves)

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve in M . We say that γ is minimizing if

$$L(\gamma) = d_g(\gamma(a), \gamma(b))$$

Definition 4.2.2 (Locally Minimizing Curves)

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve in M . We say that γ is locally minimizing if for all $t \in [a, b]$, there exists a neighbourhood $[c, d] \subseteq [a, b]$ of t such that $\gamma|_{[c, d]}$ is a minimizing curve.

Proposition 4.2.3

Let (M, g) be a Riemannian manifold. Let $\gamma : [a, b] \rightarrow M$ be a smooth curve in M . Then the following are true.

- If γ is a minimizing curve, then γ is a geodesic.
- If γ is a geodesic, then γ is a locally minimizing curve.

4.3 Geodesically Complete Riemannian Manifolds

Definition 4.3.1 (Geodesically Complete)

Proposition 4.3.2

Geodesically complete iff exp map well defined

Theorem 4.3.3 (Hopf-Rinow Theorem)

Let (M, g) be a connected Riemannian manifold. Then the following are equivalent.

- M is geodesically complete.
- M is complete as a metric space.
- Every closed and bounded set in M is compact.

Proposition 4.3.4

Let (M, g) be a connected Riemannian manifold. Let $p, q \in M$. Then there exists a minimizing geodesic from p to q .

5 Gauss-Bonnet Theorem

5.1 The Second Fundamental Form

5.2 Gauss-Bonnet Theorem

Theorem 5.2.1 (The Gauss-Bonnet Formula) Let (M, g) be an oriented smooth 2-manifold. Let γ be a positively oriented curved polygon in M and let Ω be its interior. Then

$$\int_{\Omega} K \, dA + \int_{\gamma} \kappa_N \, ds + \sum_{i=1}^k \varepsilon_i = 2\pi$$

where

- K is the Gaussian curvature of g
- dA is the Riemannian volume form
- ε_i are the exterior angles of γ
- The second integral is taken with respect to arc length

Theorem 5.2.2 (Gauss-Bonnet Theorem) Let (M, g) be a smooth compact 2-dimensional Riemannian manifold. Let K be the Gaussian curvature of M and let k_g be the geodesic curvature of ∂M . Then

$$\int_M K \, dA + \int_{\partial M} k_g \, ds = 2\pi\chi(M)$$

Corollary 5.2.3 Let (M, g) be a smooth compact 2-dimensional Riemannian manifold without boundary. Let K be the Gaussian curvature of M . Then

$$\int_M K \, dA = 2\pi\chi(M)$$

6 Differential Operators

6.1 The Riemannian Volume Form and Density

Definition 6.1.1 (The Riemannian Volume Form)

Let (M, g) be an orientable Riemannian manifold. Define the volume form dV of M to be given locally at $p \in M$ on the chart $(U, x^1, \dots, x^n)$ by

$$(dV)_p = \sqrt{\det(g_{ij}(p))} (dx^1)_p \wedge \dots \wedge (dx^n)_p$$

where g is given locally at the chart by $g = \sum_{1 \leq i, j \leq n} g_{ij}(p) dx^i \otimes dx^j$.

Let (M, g) be a compact orientable Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a compactly supported smooth function. Then we may define the integral of f over M to be the value $\int_M f dV$.

Definition 6.1.2 (The Induced Volume Form at the Boundary)

Let (M, g) be an orientable Riemannian manifold with boundary. Define the induced volume form on ∂M to be

$$dS = \iota^* dV$$

Lemma 6.1.3

Let (M, g) be an orientable Riemannian manifold with boundary. Let (U, ϕ) be a boundary chart of M . Then locally on U , the induced volume form is given by

$$dS = \sqrt{\det(g_{ij}|_{\partial M})} dx^1 \wedge \dots \wedge dx^n$$

Definition 6.1.4 (Riemannian Density)

Let (M, g) be a Riemannian manifold. Define the Riemannian density μ to be the density given at a local chart $(U, x^1, \dots, x^n)$ by

$$\mu = \sqrt{\det(g_{ij})} |dx^1 \wedge \dots \wedge dx^n|$$

Proposition 6.1.5

If M is orientable, then volume form of M is just the Riemannian density.

Definition 6.1.6 (Integrating Functions)

Let (M, g) be a Riemannian manifold. Let μ be the Riemannian density of M . Let $f \in C^\infty(M)$ such that $f\mu$ is a compactly supported density. Define the integral of f on M by

$$\int_M f \mu$$

6.2 Differential Operators

Definition 6.2.1 (The Gradient of a Function)

Let (M, g) be a Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Define the gradient of f to be

$$\text{grad}(f) = (df)^\#$$

By definition, the gradient of f is the unique section of TM such that

$$(df)_p(w) = g_p(\text{grad}(f)_p, w)$$

for all $p \in M$ and $w \in T_p M$.

Proposition 6.2.2

Let (M, g) be a Riemannian manifold. Let $(U, x^1, \dots, x^n)$ be a chart in M . Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Let g be given locally at the chart by

$$g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p$$

Let $(g^{i,j}(p)) = (g_{i,j}(p))^{-1}$ be the inverse of the matrix. Then locally at U , the gradient of f at p is given by

$$\text{grad}(f)(p) = \sum_{1 \leq i, j \leq n} \frac{\partial f}{\partial x^i} \Big|_p g^{i,j}(p) \frac{\partial}{\partial x^j} \Big|_p$$

Example 6.2.3

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. Then the gradient of f at $p \in \mathbb{R}^n$ is given by

$$\text{grad}(f)(p) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \Big|_p \frac{\partial}{\partial x_i} \Big|_p$$

Definition 6.2.4 (The Divergence of a Vector Field)

Let (M, g) be a Riemannian manifold. Let $X : M \rightarrow TM$ be a smooth vector field. Define the divergence of X to be the unique smooth function $\text{div}(X) : M \rightarrow \mathbb{R}$ such that for each $p \in M$ and $(U, x^1, \dots, x^n)$ a chart at p , we have

$$(d(X \lrcorner dV))_p = (\text{div}(X)(p))(dV)_p$$

where dV is the local top form given by $(dV)_p = \sqrt{\det(g_{i,j}(p))} (dx^1)_p \wedge \dots \wedge (dx^n)_p$ and g is given locally at the chart by $g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p$.

Proposition 6.2.5

Let (M, g) be a Riemannian manifold. Let $(U, x^1, \dots, x^n)$ be a chart in M . Let $X : M \rightarrow TM$ be a smooth vector field. Suppose that g and X are given locally at the chart by

$$g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p \quad \text{and} \quad X(p) = \sum_{i=1}^n a_i \frac{\partial}{\partial x^i} \Big|_p$$

respectively. Then locally at U , the divergence of X at $p \in U$ is given by

$$\text{div}(X)(p) = \frac{1}{\sqrt{\det(g_{i,j}(p))}} \sum_{k=1}^n \frac{\partial (a_k \circ \sqrt{\det(g_{i,j}))}}{\partial x^k} \Big|_p$$

Example 6.2.6

Let $X : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a smooth vector field whose components are given by $X_i : \mathbb{R}^n \rightarrow \mathbb{R}$. Then the divergence of X at $p \in \mathbb{R}^n$ is given by

$$\text{div}(X)(p) = \sum_{i=1}^n \frac{\partial X_i}{\partial x_i} \Big|_p$$

Definition 6.2.7 (The Laplacian of a Function)

Let (M, g) be a Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Define the Laplacian of f to be the function $\Delta f : M \rightarrow \mathbb{R}$ given by

$$\Delta f = \text{div}(\text{grad}(f))$$

Proposition 6.2.8

Let (M, g) be a Riemannian manifold. Let $f : M \rightarrow \mathbb{R}$ be a smooth function. Let $(U, x^1, \dots, x^n)$ be a chart in M . Let g be given locally at the chart by

$$g_p = \sum_{1 \leq i, j \leq n} g_{i,j}(p) (dx^i)_p \otimes (dx^j)_p$$

Let $(g^{i,j}(p)) = (g_{i,j}(p))^{-1}$ be the inverse of the matrix. Then locally at U , the laplacian of f at p is given by

$$\Delta(f)(p) = \frac{1}{\sqrt{\det(g_{i,j}(p))}} \sum_{1 \leq k, l \leq n} \frac{\partial(g^{k,l} \circ \sqrt{\det(g_{i,j})} \circ \frac{\partial f}{\partial x^l})}{\partial x^k} \Big|_p$$

Example 6.2.9

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth function. Then the laplacian of f at $p \in \mathbb{R}^n$ is given by

$$\Delta(f)(p) = \sum_{i=1}^n \frac{\partial^2 f}{\partial x_i^2} \Big|_p$$

6.3 Curl and The Case of Dimension 3**Definition 6.3.1 (The β -Operator)**

Let (M, g) be an oriented Riemannian manifold of dimension 3. Define the β -operator to be the map $\beta : TM \rightarrow \bigwedge_{i=1}^2(T^*M)$ by

$$\beta(X) = X \lrcorner dV$$

Lemma 6.3.2

Let (M, g) be an oriented Riemannian manifold of dimension 3. Then β is a smooth bundle isomorphism.

Definition 6.3.3 (The Curl of a Vector Field)

Let (M, g) be an oriented Riemannian manifold of dimension 3. Let $X : M \rightarrow TM$ be a smooth vector field. Define the curl of X to be the vector field $\text{curl}(X) : M \rightarrow TM$ given by

$$\text{curl}(X) = \beta^{-1}(d(X^\flat))$$

Proposition 6.3.4

Let (M, g) be an oriented Riemannian manifold of dimension 3. Then the following diagram is commutative:

$$\begin{array}{ccccccc} \mathcal{C}^\infty(M) & \xrightarrow{\text{grad}} & \mathfrak{X}(M) & \xrightarrow{\text{curl}} & \mathfrak{X}(M) & \xrightarrow{\text{div}} & \mathcal{C}^\infty(M) \\ \text{id} \downarrow & & \downarrow \flat & & \downarrow \beta & & \downarrow \flat \\ \Omega^0(M) & \xrightarrow{d} & \Omega^1(M) & \xrightarrow{d} & \Omega^2(M) & \xrightarrow{d} & \Omega^3(M) \end{array}$$

6.4 Classical Vector Calculus

Let M be an oriented smooth manifold of dimension n . let $p \in \partial M$. Then $T_p(\partial M)$ is an $(n-1)$ -dimensional vector subspace of $T_p(M)$. Recall that a boundary chart of M at $p \in M$ is a chart (U, ϕ) such that the following are true.

- $\phi(U) \subseteq \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n \geq 0\}$.
- $\phi^{-1}(\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n = 0\}) \subseteq \partial M$.
- $\phi(p) = 0$.

Definition 6.4.1 (The Unit Normal Vector Field)

Let (M, g) be an oriented Riemannian manifold with boundary. Define the outward point unit normal vector field to be the unique smooth section $\nu : \partial M \hookrightarrow M \rightarrow TM$ such that the following are true.

- $g_p(\nu(p), w) = 0$ for all $w \in T_p M$ and all $p \in M$.
- $g_p(\nu(p), \nu(p)) = 1$ for all $p \in M$.
- For any $p \in M$ and any boundary chart (U, ϕ) , the n th coordinate of $(\phi_*)(\nu_p)$ is strictly negative.

Lemma 6.4.2

Let (M, g) be an oriented Riemannian manifold of dimension n with boundary. Let (U, ϕ) be the boundary chart of M . Then locally at $U \cap \partial M$, the unit normal vector field is given by

$$\nu = \frac{1}{\sqrt{g^{nn}}} \sum_{j=1}^n g^{jn} \frac{\partial}{\partial x^j}$$

Lemma 6.4.3

Let (M, g) be an orientable Riemannian manifold with boundary. Then we have

$$dS = \iota^*(\nu^\flat \lrcorner dV)$$

Theorem 6.4.4 (The Gradient Theorem)

Let (M, g) be a Riemannian manifold. Let $\gamma : [0, 1] \rightarrow M$ be a smooth curve. Let $f \in C^\infty(M)$ be a smooth function. Then we have

$$\int_\gamma df = f(\gamma(1)) - f(\gamma(0))$$

Theorem 6.4.5 (The Divergence Theorem)

Let (M, g) be a compact oriented Riemannian manifold with boundary. Let $\iota : \partial M \hookrightarrow M$ be the inclusion map. Let $X \in \mathfrak{X}(M)$. Then we have

$$\int_M \operatorname{div}(X) dV_g = \int_{\partial M} g(X, \nu) dS_g$$

7 The Hodge Star Operator

7.1 The Hodge Star Operator

Definition 7.1.1 (The Hodge Star Operator)

Let (M, g) be an orientable Riemannian manifold. Let $n = \dim(M)$.

- Let $p \in M$. Define $*_p : \Lambda^k T_p^*(M) \rightarrow \Lambda^{n-k} T_p^*(M)$ to be the unique $\mathbb{C}$ -linear isomorphism such that

$$\alpha \wedge (*_p \beta) = g_p(\alpha, \beta) (dV)_p$$

where $\alpha, \beta \in \Lambda^k T_p^*(M)$ and Vol is the volume form of M associated to g , and $\langle -, - \rangle_p$ is the induced metric on $\Lambda^k(T_p^*(M))$.

- Define the Hodge star operator of M to be the map

$$* \in \Gamma(M, \text{Hom}(\Omega^k(M), \Omega^{n-k}(M)))$$

given by $p \mapsto *_p$.

Lemma 7.1.2

Let (M, g) be an orientable Riemannian manifold. Then the following are true regarding the Hodge star operator.

- $*f = f dV$ for any smooth function $f \in C^\infty(M)$.
- $**\omega = (-1)^{k(n-k)}\omega$ for any smooth k -form $\omega \in \Omega^k(M)$.

Lemma 7.1.3

Let (M, g) be an orientable Riemannian manifold. Let $X : M \rightarrow TM$ be a smooth vector field. Then the following are true.

- $X \lrcorner dV = *(X^\flat)$.
- $\text{div}(X) = (* \circ d \circ *)(X^\flat)$

Lemma 7.1.4

Let (M, g) be an orientable Riemannian manifold of dimension 3. Let $X : M \rightarrow TM$ be a smooth vector field. Then we have

$$\text{curl}(X) = (* \circ d(X^\flat))^\#$$

7.2 The L^2 -Norm

Definition 7.2.1 (The L^2 -Norm)

Let (M, g) be an orientable Riemannian manifold. Define the L^2 -norm on M to be the map $\langle -, - \rangle_{L^2} : \Omega^k(M) \times \Omega^k(M) \rightarrow \mathbb{R}$ by the formula

$$\langle \alpha, \beta \rangle_{L^2} = \int_M \langle \alpha, \beta \rangle_g dV$$

where $\langle -, - \rangle_g$ is the induced inner product on k -forms.

Lemma 7.2.2

Let (M, g) be an orientable Riemannian manifold. Then $\langle -, - \rangle_{L^2}$ defines an inner product on the compactly supported smooth k -forms on M for any $k \in \mathbb{N}$.

Lemma 7.2.3

Let (M, g) be an orientable Riemannian manifold. Let $\alpha, \beta \in \Omega^k(M)$ be compactly supported smooth k -forms. Then we have

$$\langle \alpha, \beta \rangle_{L^2} = \int_M \alpha \wedge * \beta$$

7.3 The Adjoint Differential Operator

Definition 7.3.1 (The Adjoint Differential Operator)

Let (M, g) be an orientable Riemannian manifold. Define the operator $d^* : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$ by

$$d^*(\omega) = (-1)^{n(k+1)+1}(* \circ d \circ *)(\omega)$$

for $\omega \in \Omega^k(M)$. Also define $d^*(\omega) = 0$ for $\omega \in \Omega^0(M)$ by convention.

Lemma 7.3.2

Let (M, g) be an orientable Riemannian manifold. Then we have $d^* \circ d^* = 0$.

Proposition 7.3.3

Let (M, g) be a compact orientable Riemannian manifold. Then we have

$$\langle d^*\omega, \eta \rangle_{L^2} = \langle \omega, d\eta \rangle_{L^2}$$

for any $\omega \in \Omega^{k+1}(M)$ and $\eta \in \Omega^k(M)$.

Definition 7.3.4 (The Laplace-Beltrami Operator)

Let M be an orientable Riemannian manifold. Define the Laplace-Beltrami operator on M to be the map $\Delta : \Omega^k(M) \rightarrow \Omega^k(M)$ given by

$$\Delta = d^* \circ d + d \circ d^*$$

Lemma 7.3.5

Let M be an orientable Riemannian manifold. Then the following are true regarding the Laplace-Beltrami operator.

- $* \circ \Delta = \Delta \circ *$,
- $\langle \Delta\alpha, \beta \rangle_{L^2} = \langle \alpha, \Delta\beta \rangle_{L^2}$ for any compactly supported differential forms $\alpha, \beta \in \Omega^k(M)$.

7.4 The Hodge Theorem for Riemannian Manifolds

Definition 7.4.1 (Harmonic Forms)

Let M be an orientable Riemannian manifold. Let $\alpha \in \Omega^k(M)$. We say that α is harmonic if $\alpha \in \ker(\Delta)$.

Lemma 7.4.2

Let M be an orientable Riemannian manifold. Let $\alpha \in \Omega^k(M)$. Then α is harmonic if and only if $d(\alpha) = 0$ and $d^*(\alpha) = 0$.

Definition 7.4.3 (The Space of Harmonic Forms)

Let M be an orientable Riemannian manifold. Define the space of harmonic k -forms to be

$$\mathcal{H}^k(M) = \ker(\Delta)$$

Theorem 7.4.4 (Hodge Theorem)

Let M be a compact orientable Riemannian manifold. Then the following are true.

- $\mathcal{H}^k$ is finite dimensional for all $k \in \mathbb{N}$.
- There is a decomposition

$$\Omega^k(M) \cong \mathcal{H}^k(M) \oplus \text{im}(\Delta)$$

Corollary 7.4.5

Let M be an orientable Riemannian manifold. Then the quotient map induces an isomorphism

$$\mathcal{H}^k(M) \cong H_{\text{dR}}^k(M)$$

for all $k \in \mathbb{N}$.

8 Holonomy

8.1 Holonomy Groups

Definition 8.1.1 (Holonomy Groups)

Let (M, g) be a Riemannian manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Let $x \in M$. Define the holonomy group of E at x with respect to ∇ to be

$$\text{Hol}_x(\nabla) = \{P_\gamma \mid \gamma(0) = \gamma(1) = x\} \subseteq \text{GL}(E_x)$$

Lemma 8.1.2

Let (M, g) be a Riemannian manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Let $x \in M$. Then $\text{Hol}_x(\nabla)$ is a Lie subgroup of $\text{GL}(E_x)$.

Proposition 8.1.3

Let (M, g) be a Riemannian manifold. Let $p : E \rightarrow M$ be a smooth vector bundle. Let ∇ be a connection on E . Then the following are true.

- Let $\gamma : [0, 1] \rightarrow M$ be a path. Then we have

$$\text{Hol}_{\gamma(1)}(\nabla) = P_\gamma \text{Hol}_{\gamma(0)}(\nabla) P_\gamma^{-1}$$

- If M is simply connected, then $\text{Hol}_x(\nabla)$ is connected for all $x \in M$.

So for simply connected spaces, the holonomy groups are independent of base point up to isomorphism. But the specific embedding into $\text{GL}(n, F)$ does depend on the base point.

Definition 8.1.4

Let M be a simply connected Riemannian manifold. Let $p : E \rightarrow M$ be a smooth vector bundle over $\mathbb{R}$. Let ∇ be a connection on E . Define the holonomy group of E with respect to ∇ to be

$$\text{Hol}(\nabla) = \text{Hol}_x(\nabla)$$

for any base point $x \in M$.

Proposition 8.1.5

Let (M, g) be a Riemannian manifold. Let $p \in M$. Let $\phi : T_p M \rightarrow \mathbb{R}^n$ be an $\mathbb{R}$ -linear isomorphism. Then we have

$$\text{GL}(\phi)(\text{Hol}_p(\nabla^g)) \leq O(n)$$

Proposition 8.1.6

Let (M, g) and (N, h) be Riemannian manifolds. Let $p \in M$ and $q \in N$. Then we have

$$\text{Hol}_{(p,q)}(\nabla^{g \times h}) = \text{Hol}_p(\nabla^g) \times \text{Hol}_q(\nabla^h)$$

Proposition 8.1.7

Let (M, g) be a Riemannian manifold. Let $p \in M$. If M is irreducible, then the representation $\text{Hol}_p(\nabla^g) \rightarrow \text{GL}(T_p M)$ is irreducible.

Theorem 8.1.8 (Berger)

Let (M, g) be a Riemannian manifold of dimension n . Suppose that M is simply connected, irreducible and not symmetric. Let $p \in M$. Then exactly one of the following cases hold.

- $\text{Hol}_p(\nabla^g) \cong SO(n)$.
- $n = 2m$ for $m \geq 2$ and $\text{Hol}_p(\nabla^g) \cong U(m) \leq SO(n)$.
- $n = 2m$ for $m \geq 2$ and $\text{Hol}_p(\nabla^g) \cong SU(m) \leq SO(n)$.
- $n = 4m$ for $m \geq 2$ and $\text{Hol}_p(\nabla^g) \cong SP(m) \leq SO(n)$.

- $n = 4m$ for $m \geq 2$ and $\text{Hol}_p(\nabla^g) \cong SP(m) \times SP(1) \leq SO(n)$.
- $n = 7$ and $\text{Hol}_p(\nabla^g) \cong G_2 \leq SO(7)$.
- $n = 8$ and $\text{Hol}_p(\nabla^g) \cong \text{Spin}(7) \leq SO(8)$.

9 Pseudo-Riemannian Manifolds

9.1 Basic Definitions

In linear algebra, inner products are symmetric and positive-definite bilinear forms. This implies that inner products are non-degenerate. We may relax the positive-definiteness condition with non-degeneracy, and such bilinear forms are closely related to quadratic forms. Similarly, we may relax the Riemannian metric to such bilinear forms.

Definition 9.1.1 (Pseudo-Riemannian Metrics)

Let M be a smooth manifold. A pseudo-Riemannian metric on M is a smooth section

$$g \in \mathcal{C}^\infty((TM \otimes TM)^*)$$

such that $g_p : T_p M \times T_p M \rightarrow \mathbb{R}$ is a symmetric and non-degenerate bilinear form for all $p \in M$.

Definition 9.1.2 (Pseudo-Riemannian Manifolds)

A pseudo-Riemannian manifold is a smooth manifold M together with a pseudo-Riemannian metric g on M .

Proposition 9.1.3

Let (M, g) be a pseudo-Riemannian manifold. Let $p \in M$. Let $(U, x^1, \dots, x^n)$ be a chart at p . Then g_p as an element of $T_p(M)^* \otimes T_p(M)^*$ is given locally on the chart by

$$g = \sum_{1 \leq i, j \leq n} g_{ij} dx^i \otimes dx^j$$

where $g_{ij} : U \rightarrow \mathbb{R}$ are smooth functions such that $g_{ij}(p) = g_p \left(\left. \frac{\partial}{\partial x^i} \right|_p, \left. \frac{\partial}{\partial x^j} \right|_p \right)$.

Definition 9.1.4 (Signature of the Metric)

Let (M, g) be a pseudo-Riemannian manifold. Define the signature of g to be the signature of g_p as a quadratic form.

Not every smooth manifold admits a metric with signature (p, q) whenever $q \neq 0$.

9.2 The Musical Isomorphisms

9.3 The Levi-Civita Connection

9.4 Curvature

9.5 Calculus on Manifolds

9.6 Hodge Star Operator